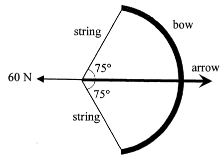
ch10 Projectile Motion

Q5

5. (a) A bow and arrow is a kind of projectile weapon. The string of a bow is drawn taut by a hunter with a force of 60 N and an arrow of mass 0.2 kg is held stationary as shown in Figure 5.1.

Figure 5.1



(i) Find the tension of the string. Neglect the weight of the arrow. (2 marks)

(ii) Estimate the energy stored in the taut string if the initial speed of the arrow is 45 m s^{-1} when released. Assume that the bow is rigid and neglect the mass of the string. (2 marks)

Answers written in the margins will not be marked.

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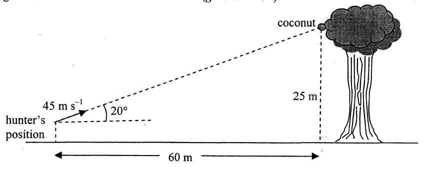
*(b) The hunter stands at about 60 m away from a tree as shown in Figure 5.2. He uses the bow to release the arrow in order to shoot a coconut held by a monkey (not shown in the figure) in the tree. The coconut is at a height of 25 m from the ground. The hunter aims directly at the coconut and the arrow leaves the bow at a speed of 45 m s^{-1} making an angle of 20° to the horizontal. At the moment the hunter releases the arrow, the monkey drops the coconut such that it falls vertically from rest. Neglect air resistance and the arrow's size. (g = 9.81 m s^{-2})

Figure 5.2



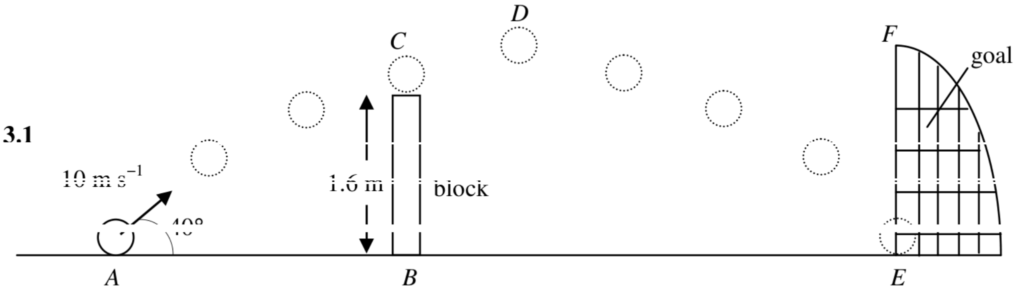
(i) Find the time taken for the arrow to hit the coconut. (2 marks)

(ii) Find the height of the coconut from the ground at the moment the arrow hits it. (2 marks)

Answers written in the margins will not be marked.

3. A ball is kicked and moves with an initial velocity of 10 m s^{-1} at an angle of 40° to the horizontal. The ball then just passes a block of height 1.6 m, reaching the highest point D, and finally hits the ground at E as shown in Figure 3.1. (Neglect air resistance and the size of the ball.)

Figure 3.1



(a) Draw an arrow to indicate the direction of acceleration of the ball at C. (1 mark)

*(b) For a projectile of initial velocity u that makes an angle θ with the horizontal, show that its horizontal range is given by $\dfrac{u^2 \sin 2\theta}{g}$. Hence, or otherwise, find another angle of projection such that the ball can still reach E with the same initial speed of 10 m s^{-1}.
(Given: $2\sin\theta\cos\theta = \sin 2\theta$) (4 marks)

(c) Calculate the speed of the ball at C. (2 marks)